COMPARATIVE REVIEW OF THREE RESEARCH PAPERS

Student Skills and Course Recommendation System

Purpose: Summarize three research papers relevant to student skill analysis and personalized course recommendation, compare their scope and contributions, and present a combined conclusion for the selected SPI topic.

Research papers reviewed

Domain-specific NLP system to support learning path and curriculum design at tech universities (2021/2022).

A Recommender System of Postgraduate Courses Based on Soft Skills: A Psychometric-inspired Approach (2025).

Personalised Course Recommender: Linking Learning Objectives and Career Goals through Competencies (2024).

Overview: These papers cover three complementary aspects: extracting technical skills from text and recommending learning paths, matching courses to soft-skill development needs, and connecting course learning objectives with career competencies.

Summary of Research Paper 1

Title: Domain-specific NLP system to support learning path and curriculum design at tech universities (Nhi N.Y. Vo, Quang T. Vu, Nam H. Vu, Tu A. Vu, Bang D. Mach and Guandong Xu; Computers and Education: Artificial Intelligence, 2022; first published online in 2021).

Focus and objective. Develops an NLP-based course recommendation system for computer science and IT students. It identifies technical skills from course and job descriptions and uses them to support personalized learning and curriculum decisions.

Main contribution. Combines a domain-specific named-entity recognition model (CSIT-NER) with a hybrid course recommendation model (CSIT-CRS). It can recommend university courses, career paths/job listings, and online courses aligned with industry skills. The authors report feedback from 201 student and faculty participants.

Limitations and interpretation. The system focuses on CS/IT and depends on the quality and coverage of technical text and skill data. Suitability for other study domains would need additional validation.

Link: https://doi.org/10.1016/j.caeai.2021.100042

Summary of Research Paper 2

Title: A Recommender System of Postgraduate Courses Based on Soft Skills: A Psychometric-inspired Approach (Luis Alberto Pinos Ullauri, Alexis Lebis, Abir Karami, Mathieu Vermeulen, Anthony Fleury and colleagues; International Journal of Artificial Intelligence in Education, 2025).

Focus and objective. Proposes a course recommender to address the gap between students' actual soft-skill proficiency and the proficiency expected by employers. It recommends combinations of courses that may help students develop multiple soft skills.

Main contribution. Uses psychometric modelling to estimate students' soft-skill proficiency and a genetic-algorithm framework to evaluate course sets across ten soft skills, including problem-solving and leadership. The paper reports that recommendations had higher average fitness than the courses students actually took.

Limitations and interpretation. The framework is illustrated using data from three cohorts at one French higher-education institution, so results may not generalize directly to all institutions or student populations.

Link: https://doi.org/10.1007/s40593-025-00463-z

Summary of Research Paper 2

Title: Personalised Course Recommender: Linking Learning Objectives and Career Goals through Competencies (Nils Beutling and Maja Spahic-Bogdanovic; Proceedings of the AAAI Spring Symposium Series, 2024, pp. 72–81).

Focus and objective. Presents a knowledge-based recommender for information-systems education, connecting students' career goals to course learning objectives and the competencies required for particular jobs.

Main contribution. Uses the ESCO skills/competencies/occupations ontology, course descriptions and a large language model to extract learning objectives and map them to job-related competencies. Course relevance is estimated from how many relevant job skills a course supports, with explanations to help students decide.

Limitations and interpretation. The approach is demonstrated in information-systems education and depends on course descriptions, competency mappings and ontology coverage.

Link: https://doi.org/10.1609/aaaiss.v3i1.31185

Comparison of All Three Research Papers

Combined Conclusion

Together, these papers suggest that a student-focused system can combine technical-skill extraction, soft-skill assessment and career-competency matching. A practical system could collect a student's current skills and interests, identify gaps against a chosen career goal, and recommend courses that address those gaps. Recommendations should include understandable reasons and be evaluated for relevance, data quality and fairness. Since each paper is tested in a particular domain or dataset, a new system should be evaluated with its intended student population.

Objectives

To study NLP and recommendation techniques used to identify student skills and suggest relevant courses.

To understand how technical and soft skills can be mapped to course learning outcomes.

To compare NLP-based, psychometric and knowledge-based recommendation approaches.

To identify skill gaps by comparing current competencies with skills required for selected career goals.

To highlight explainable recommendations and evaluation using student/course data.

References

Vo, N. N. Y., Vu, Q. T., Vu, N. H., Vu, T. A., Mach, B. D., & Xu, G. (2022). Domain-specific NLP system to support learning path and curriculum design at tech universities. Computers and Education: Artificial Intelligence, 3, 100042. https://doi.org/10.1016/j.caeai.2021.100042

Pinos Ullauri, L. A., et al. (2025). A Recommender System of Postgraduate Courses Based on Soft Skills: A Psychometric-inspired Approach. International Journal of Artificial Intelligence in Education, 35, 2117–2153. https://doi.org/10.1007/s40593-025-00463-z

Beutling, N., & Spahic-Bogdanovic, M. (2024). Personalised Course Recommender: Linking Learning Objectives and Career Goals through Competencies. Proceedings of the AAAI Spring Symposium Series, 3(1), 72–81. https://doi.org/10.1609/aaaiss.v3i1.31185

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Comparison point	Paper 1: NLP learning paths	Paper 2: Soft-skill courses	Paper 3: Career competencies
Main topic	Technical-skill extraction and course recommendations	Courses to improve soft skills	Courses mapped to career goals and job skills

Comparison point	Paper 1: NLP learning paths	Paper 2: Soft-skill courses	Paper 3: Career competencies
Research type	Journal article (2022; online 2021)	Journal article (2025)	AAAI symposium paper (2024)
Main techniques	NLP, named-entity recognition, hybrid recommender	Psychometric modelling, genetic algorithm	Knowledge-based recommender, ESCO ontology, LLM
Key strength	Links technical skills with courses and jobs	Considers multiple soft skills in course selection	Explains course relevance using career competencies
Key limitation	CS/IT focus and dependence on skill-text data	Evidence from three cohorts at one institution	Depends on course descriptions and competency mapping